

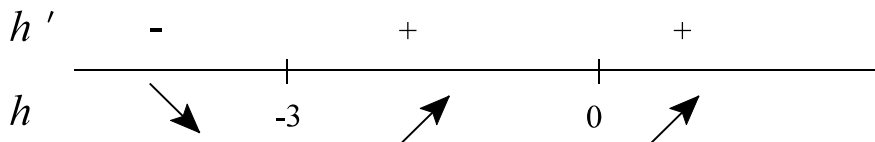
$$h(x) = 3x^4 + 12x^3 - 2$$

a. Domain = $\mathbb{R}$

b. Intercepts $(0, -2)$, $(-4.0103, 0)$, $(0.5280, 0)$

c. $h'(x) = 12x^3 + 36x^2 \Rightarrow$ C.N. $x = -3, x = 0$

d.



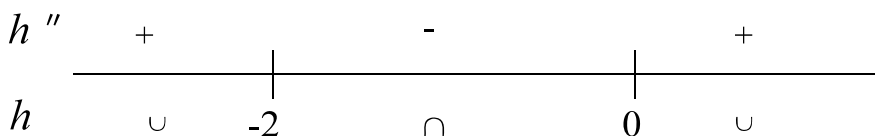
h is decreasing on $(-\infty, -3)$

h is increasing on $(-3, 0) \cup (0, \infty)$

e. $P(-3, -83)$ is a relative minimum on the graph

f. $h''(x) = 36x^2 + 72x \Rightarrow$ 2nd C.N. $x = -2, x = 0$

g.



h is concave up on $(-\infty, -2) \cup (0, \infty)$

h is concave down on $(-2, 0)$

h. Inflection points $P(-2, -50)$, $P(0, -2)$

i. -6, 1294

-5, 373

-4, -2

-3, -83

-2, -50

-1, -11

0, -2

1, 13

2, 142

3, 565

4, 1534

j.

